Class 2 HW - Student

```
# fit_data: fit a workflow to a full dataset using a formula and parsnip spec
fit_data <- function(formula, model, data, ...) {
  wf <- workflows::workflow() |>
    workflows::add_formula(formula) |>
    workflows::add_model(model)
  fit(wf, data, ...)
}

# fit_split: fit a workflow and evaluate on a held-out split (last_fit pattern)
fit_split <- function(formula, model, split, ...) {
  wf <- workflows::workflow() |>
    workflows::add_model(model) |>
    workflows::add_formula(
      formula,
      blueprint = hardhat::default_formula_blueprint(
        indicators = FALSE, allow_novel_levels = TRUE
      )
    )
  tune::last_fit(wf, split, ...)
}
```

1

Create a `parsnip` specification for a linear regression model.

```
library(tidymodels)
```

```
## Warning: package 'tidymodels' was built under R version 4.5.3
```

```
## -- Attaching packages --------------------------------------- tidymodels 1.5.0 --
```

```
## v broom        1.0.12     v recipes      1.3.3
## v dials        1.4.3      v rsample      1.3.2
## v dplyr        1.2.0      v tailor       0.1.0
## v ggplot2      4.0.3      v tidyr        1.3.2
## v infer        1.1.0      v tune         2.1.0
## v modeldata    1.5.1      v workflows    1.3.0
## v parsnip      1.6.0      v workflowsets 1.1.1
## v purrr        1.2.1      v yardstick    1.4.0
```

```
## Warning: package 'dials' was built under R version 4.5.3
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
## Warning: package 'infer' was built under R version 4.5.3

## Warning: package 'modeldata' was built under R version 4.5.3

## Warning: package 'parsnip' was built under R version 4.5.3

## Warning: package 'recipes' was built under R version 4.5.3

## Warning: package 'rsample' was built under R version 4.5.3

## Warning: package 'tailor' was built under R version 4.5.3

## Warning: package 'tune' was built under R version 4.5.3

## Warning: package 'workflows' was built under R version 4.5.3

## Warning: package 'workflowsets' was built under R version 4.5.3

## Warning: package 'yardstick' was built under R version 4.5.3

## -- Conflicts --------------------------------------- tidymodels_conflicts() --
## x purrr::discard() masks scales::discard()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x recipes::step()  masks stats::step()
```

```
library(MASS)
```

```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select
```

```
library(ISLR)
```

```
## Warning: package 'ISLR' was built under R version 4.5.3
```

HW code

```
lm_spec <- linear_reg() %>%
  set_mode("regression") %>%
  set_engine("lm")
```

2

Once we have the specification we can `fit` it by supplying a formula expression and the data we want to fit the model on. The formula is written on the form `y ~ x` where `y` is the name of the response and `x` is the name of the predictors. The names used in the formula should match the names of the variables in the data set passed to `data`.

Use `lm_spec` to fit the model of `medv` predicted by every predictor variable. Hint: you can use "." to specify every predictor.

HW code

```
lm_fit <- lm_spec %>%
  fit(medv ~ ., data = Boston)
lm_fit
```

```
## parsnip model object
##
##
## Call:
## stats::lm(formula = medv ~ ., data = data)
##
## Coefficients:
## (Intercept)         crim           zn        indus         chas          nox
##   3.646e+01   -1.080e-01    4.642e-02    2.056e-02    2.687e+00   -1.777e+01
##          rm          age          dis          rad          tax      ptratio
##   3.810e+00    6.922e-04   -1.476e+00    3.060e-01   -1.233e-02   -9.527e-01
##       black        lstat
##   9.312e-03   -5.248e-01
```

3

Get a summary of your model using `pluck` and `summary`

HW code

```
lm_fit %>%
  pluck("fit") %>%
  summary()
```

```
##
## Call:
## stats::lm(formula = medv ~ ., data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -15.595  -2.730  -0.518   1.777  26.199
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.646e+01  5.103e+00   7.144 3.28e-12 ***
## crim        -1.080e-01  3.286e-02  -3.287 0.001087 **
## zn           4.642e-02  1.373e-02   3.382 0.000778 ***
## indus        2.056e-02  6.150e-02   0.334 0.738288
## chas         2.687e+00  8.616e-01   3.118 0.001925 **
## nox         -1.777e+01  3.820e+00  -4.651 4.25e-06 ***
## rm           3.810e+00  4.179e-01   9.116  < 2e-16 ***
## age          6.922e-04  1.321e-02   0.052 0.958229
## dis         -1.476e+00  1.995e-01  -7.398 6.01e-13 ***
## rad          3.060e-01  6.635e-02   4.613 5.07e-06 ***
## tax         -1.233e-02  3.760e-03  -3.280 0.001112 **
## ptratio     -9.527e-01  1.308e-01  -7.283 1.31e-12 ***
## black        9.312e-03  2.686e-03   3.467 0.000573 ***
## lstat       -5.248e-01  5.072e-02 -10.347  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.745 on 492 degrees of freedom
## Multiple R-squared:  0.7406, Adjusted R-squared:  0.7338
## F-statistic: 108.1 on 13 and 492 DF,  p-value: < 2.2e-16
```

4

Take a look at `lm_fit` with `tidy`

HW Code

```
tidy(lm_fit)
```

```
## # A tibble: 14 x 5
##    term          estimate std.error statistic  p.value
##    <chr>            <dbl>     <dbl>     <dbl>    <dbl>
##  1 (Intercept)  36.5        5.10       7.14   3.28e-12
##  2 crim         -0.108      0.0329    -3.29   1.09e- 3
##  3 zn            0.0464     0.0137     3.38   7.78e- 4
##  4 indus         0.0206     0.0615     0.334  7.38e- 1
##  5 chas          2.69       0.862      3.12   1.93e- 3
##  6 nox         -17.8        3.82      -4.65   4.25e- 6
##  7 rm            3.81       0.418      9.12   1.98e-18
##  8 age           0.000692   0.0132     0.0524 9.58e- 1
##  9 dis          -1.48       0.199     -7.40   6.01e-13
## 10 rad           0.306      0.0663     4.61   5.07e- 6
## 11 tax          -0.0123     0.00376   -3.28   1.11e- 3
## 12 ptratio      -0.953      0.131     -7.28   1.31e-12
## 13 black         0.00931    0.00269    3.47   5.73e- 4
## 14 lstat        -0.525      0.0507   -10.3    7.78e-23
```

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Extract the model statistics using `glance`

#HW code

```
glance(lm_fit)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic   p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>     <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.741         0.734  4.75      108. 6.72e-135    13 -1499. 3028. 3091.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

6

Get the predicted `medv` values from your model using `predict`

#HW code

```
predict(lm_fit, new_data = Boston)
```

```
## # A tibble: 506 x 1
##    .pred
##    <dbl>
##  1  30.0
##  2  25.0
##  3  30.6
##  4  28.6
##  5  27.9
##  6  25.3
##  7  23.0
##  8  19.5
##  9  11.5
## 10  18.9
## # i 496 more rows
```

7

Bind the predicted columns to your existing data

#HW code

```
bind_cols(
  predict(lm_fit, new_data = Boston),
  Boston
) %>%
  dplyr::select(medv, .pred)
```

```
## # A tibble: 506 x 2
##     medv .pred
##    <dbl> <dbl>
##  1  24    30.0
##  2  21.6  25.0
##  3  34.7  30.6
##  4  33.4  28.6
##  5  36.2  27.9
##  6  28.7  25.3
##  7  22.9  23.0
##  8  27.1  19.5
##  9  16.5  11.5
## 10  18.9  18.9
## # i 496 more rows
```

8

Now, make things easier by just using the `augment` function to do this.

#HW code

```
augment(lm_fit, new_data = Boston)
```

```
## # A tibble: 506 x 16
##    .pred  .resid    crim    zn indus  chas   nox    rm   age   dis   rad   tax
##    <dbl>   <dbl>   <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>
##  1  30.0 -6.00   0.00632  18    2.31     0 0.538  6.58  65.2  4.09     1   296
##  2  25.0 -3.43   0.0273    0    7.07     0 0.469  6.42  78.9  4.97     2   242
##  3  30.6  4.13   0.0273    0    7.07     0 0.469  7.18  61.1  4.97     2   242
##  4  28.6  4.79   0.0324    0    2.18     0 0.458  7.00  45.8  6.06     3   222
##  5  27.9  8.26   0.0690    0    2.18     0 0.458  7.15  54.2  6.06     3   222
##  6  25.3  3.44   0.0298    0    2.18     0 0.458  6.43  58.7  6.06     3   222
##  7  23.0 -0.102  0.0883   12.5  7.87     0 0.524  6.01  66.6  5.56     5   311
##  8  19.5  7.56   0.145    12.5  7.87     0 0.524  6.17  96.1  5.95     5   311
##  9  11.5  4.98   0.211    12.5  7.87     0 0.524  5.63 100    6.08     5   311
## 10  18.9 -0.0203 0.170    12.5  7.87     0 0.524  6.00  85.9  6.59     5   311
## # i 496 more rows
## # i 4 more variables: ptratio <dbl>, black <dbl>, lstat <dbl>, medv <dbl>
```

9

Focus specifically on the median value and the `.pred`, then you can select those two columns

#HW code

```
augment(lm_fit, new_data = Boston) %>%
  dplyr::select(medv, .pred)
```

```
## # A tibble: 506 x 2
##     medv .pred
##    <dbl> <dbl>
```

```
##  1  24    30.0
##  2  21.6  25.0
##  3  34.7  30.6
##  4  33.4  28.6
##  5  36.2  27.9
##  6  28.7  25.3
##  7  22.9  23.0
##  8  27.1  19.5
##  9  16.5  11.5
## 10  18.9  18.9
## # i 496 more rows
```

10

Create a `recipe` with an interaction step between lstat and age

#HW code

```
rec_spec <- recipe(medv ~ ., data = Boston) %>%
  step_interact(~ lstat:age)
```

11

Create a `workflow` and add your lm_spec model and your rec_spec recipe.

#HW code

```
lm_wf <- workflow() %>%
  add_model(lm_spec) %>%
  add_recipe(rec_spec)
```

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Fit your `workflow`.

#HW code

```
lm_wf %>% fit(Boston)
```

```
## == Workflow [trained] ==========================================================
## Preprocessor: Recipe
## Model: linear_reg()
##
## -- Preprocessor ----------------------------------------------------------------
## 1 Recipe Step
##
## * step_interact()
##
## -- Model -----------------------------------------------------------------------
```

```
##
## Call:
## stats::lm(formula = ..y ~ ., data = data)
##
## Coefficients:
## (Intercept)         crim           zn        indus         chas          nox
##   38.034385    -0.111723     0.043109     0.018945     2.709592   -18.026578
##          rm          age          dis          rad          tax      ptratio
##    3.759388    -0.010315    -1.475346     0.307676    -0.012335    -0.958239
##       black        lstat  lstat_x_age
##    0.009187    -0.635696     0.001257
```